

Challenge (Habanero)

Q1. If $|x + 2| = 7$, then x is?

- (A) -9
- (B) 5
- (C) -5 or 9
- (D) 3
- (E) -3

Q2. A hiker's distance from the campsite after t hours is given by $|t - 5|$. How far is the hiker from the campsite after 8 hours?

- (A) 3 miles
- (B) 5 miles
- (C) 7 miles
- (D) 9 miles
- (E) 11 miles

Q3. Solve the equation $|2x - 3| = 11$.

- (A) 7 or -4
- (B) -7 or 4
- (C) 7 or 2
- (D) -2 or 7
- (E) 3 or -5

Q4. A car's fuel consumption is given by $|x - 40| + 10$, where x is the speed in miles per hour. What is the fuel consumption when the car is traveling at 50 miles per hour?

- (A) 20
- (B) 15
- (C) 10
- (D) 25
- (E) 30

Q5. The population of a certain animal species is given by $|t - 10| + 50$, where t is the number of years since the species was first observed. What is the population after 12 years?

- (A) 62
- (B) 58
- (C) 52
- (D) 60
- (E) 55

Q6. Solve the equation $|x + 2| = |3x - 1|$.

- (A) $-\frac{3}{2}$
- (A) $-\frac{3}{2}$
- (A) $-\frac{1}{2}$
- (A) $-\frac{1}{2}$
- (A) 1

Q7. If $|x - 2| = |x + 2|$, then x is?

- (A) 0
- (B) 2
- (C) -2
- (D) 1
- (E) -1

Q8. Solve the equation $|2x + 1| = |x - 3|$.

- (A) -1
- (B) 1
- (C) -2
- (D) 2
- (E) -4

Open Ended Questions (FRQ)

Q9. Solve the equation $|x^2 - 4| = 9$ and check for extraneous solutions.

Q10. A group of hikers are planning a trip to a remote area. The distance from the trailhead to the campsite is given by $|x - 10| + 5$, where x is the number of hours hiked. If the group wants to hike for 8 hours, how far will they be from the campsite? Show your work and explain your reasoning.

Chef's Tips

- Emphasize the importance of checking for extraneous solutions.
- Encourage students to use graphs to visualize the solutions to absolute value equations.
- Remind students to consider the context of the problem when solving absolute value equations.